



Manifold for T1D – DRI Long-Horizon Transplant Analytics

Islet-transplant registry causal integration and BioHub translational prioritization

The Diabetes Research Institute holds an asset no other institution can replicate: the **longest continuous programmatic record of clinical islet transplantation in the world**, paired with the active **BioHub** translational program. That record — donor quality, recipient HLA and autoantibody profile, immunosuppression regimen, longitudinal C-peptide AUC, and long-horizon graft outcome — is a substrate for causal inference at a scale and horizon unmatched elsewhere. **Manifold** configures three engine capabilities to the domain: causal discovery on the registry's covariate structure, criticality estimators (long-horizon survival, HTE learners, mixed-effects decay, change-point) feeding a composite fragility score (ΩF), and Pearl do-calculus with interdiction for principled regimen and BioHub-variant prioritization.

The DRI record demonstrates that islet replacement can produce insulin-independence. The outstanding question is which donor, recipient, regimen, and delivery-architecture combinations sustain the outcome over a decade — a problem of causal inference on a uniquely long-horizon dataset.

ΩF PILLAR MAPPING – DRI ISLET-TRANSPLANT PHYSIOLOGY

ΩF PILLAR	DRI REGISTRY READING
I — Interaction	Donor quality × recipient HLA × immunosuppression × autoantibody interactions; BioHub substrate × host-tissue
R — Reserve	Post-transplant functional graft mass (C-peptide AUC on MMTT); residual graft mass at long horizons
J — Junction	Allo- and auto-immunity coupling with metabolic response; BioHub scaffold × host-immune coupling
C — Cascade	Graft-attribution cascades: allorejection, autoimmunity recurrence, fibrotic-response cascades in BioHub substrates
T — Temporal	Hazard of graft attrition at 1, 5, 10, and 15-year horizons; time-to-insulin-dependence return

PROPOSED PARTNERSHIP

Partnership with a Diabetes Research Institute principal investigator at the University of Miami Miller School of Medicine is sought for:

1. Retrospective DRI islet-transplant registry and BioHub data access under the DRI's existing Data Use Agreement and IRB.
2. Clinical co-authorship on the long-horizon graft-survival and regimen-interdiction benchmark manuscript (twelve to fourteen weeks from data access).
3. Co-application to Breakthrough T1D Cures, Helmsley Restoring Insulin Production, or NIH NIDDK for the Phase 2 prospective arm.

Apex Analytica contributes the causal-inference engine, the ΩF framework, and the engineering team. Full brief: M-T1D-02-DRIMIAMI .